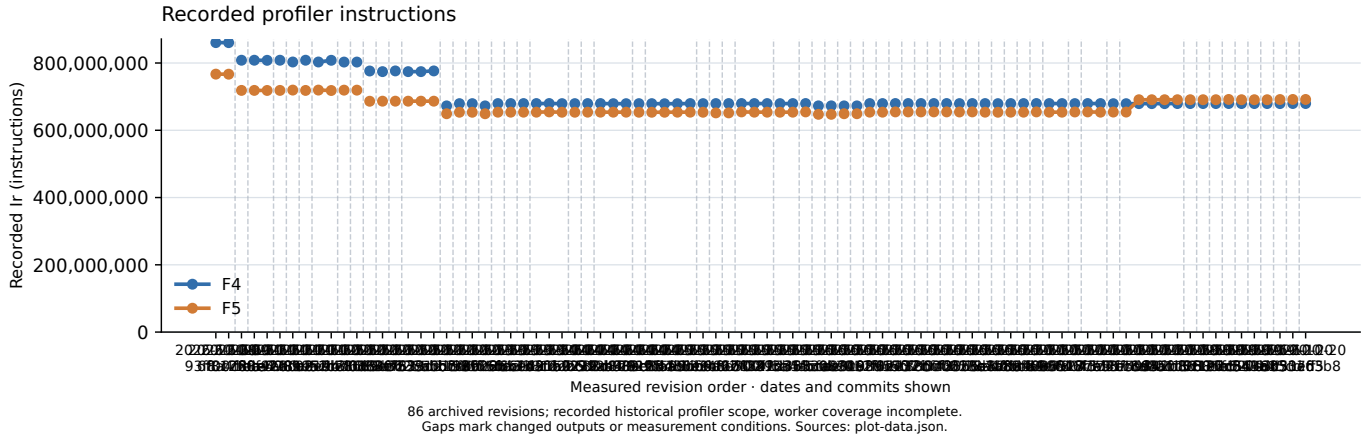
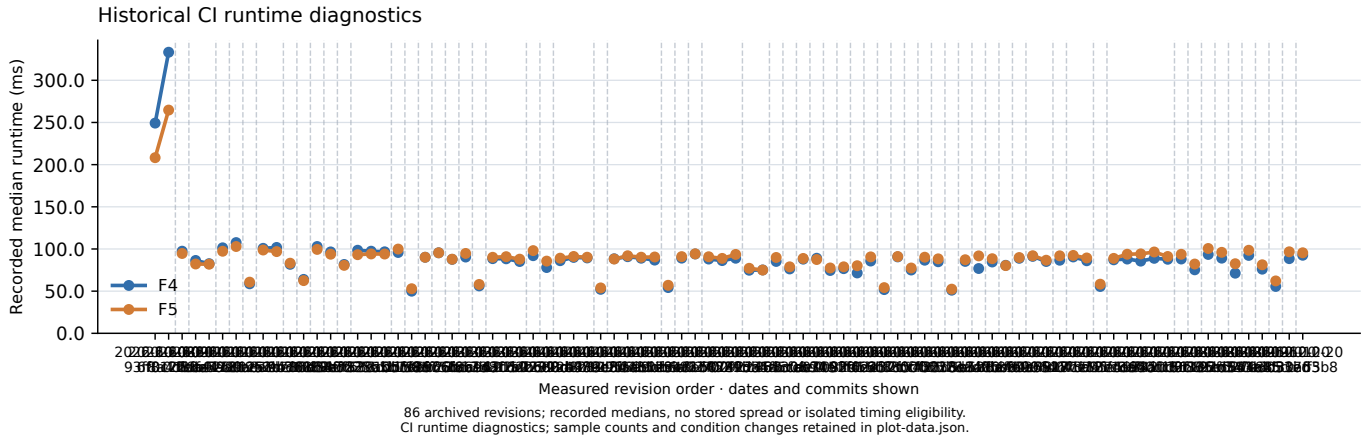
Boolean matrix step: F4 and F5

One shared K_1 / 2^23 m=2 system, 22 Boolean variables, maximum degree 4. Matrix construction, reduction and readback; F5 includes its criterion.





UTC date / revision	F4 Ir	F5 Ir	F4 ms	F5 ms
2026-10-08 / 93ff0c72	860,633,889	766,940,469	249.241	208.250
2026-10-09 / 6f81260f	860,555,567	766,776,087	333.279	264.687
2026-10-11 / 8a41eca7	808,263,705	718,576,235	97.380	94.658
2026-10-11 / 0cfba411	808,263,705	718,578,523	86.390	82.214
2026-10-11 / f88e5909	807,989,935	718,391,055	82.597	81.948
2026-10-11 / fb926c02	808,263,705	718,576,189	101.379	97.323
2026-10-11 / e478b952	803,017,657	719,516,063	107.585	102.927
2026-10-11 / aa0cee7b	808,281,869	718,492,761	58.723	60.673
2026-10-12 / 81ca1c78	803,015,877	719,509,003	100.762	98.775
2026-10-12 / a8194d61	808,072,539	718,282,030	101.827	96.904
2026-10-12 / 5cbe16d5	802,824,065	719,316,911	81.784	83.144
2026-10-12 / eb70862e	802,824,065	719,316,876	63.944	62.505
2026-10-12 / 31088bfc	776,242,363	686,315,894	102.929	99.489
2026-10-12 / 0d34f782	774,379,185	686,521,131	96.521	93.790
2026-10-12 / 89e3233c	776,490,367	686,543,371	81.628	80.581
2026-10-12 / 80f5391d	774,379,185	686,519,896	98.501	93.239
2026-10-12 / c62555b7	774,379,185	686,519,896	97.360	94.117
2026-10-12 / 633d2158	776,467,358	686,512,000	96.637	94.008
2026-10-12 / ca0d5f86	672,500,776	649,174,334	95.848	99.895
2026-10-12 / b5cc6ffc	679,242,389	653,622,849	49.965	52.882
2026-10-13 / bf92906b	679,242,389	653,622,849	90.163	90.201
2026-10-13 / d85756aa	672,486,486	649,160,075	95.497	95.480
2026-10-13 / ab28b57b	679,449,939	653,823,121	87.697	87.878
2026-10-13 / 56f66e43	679,449,939	653,828,331	90.448	94.715
2026-10-13 / 28b1daf2	679,449,939	653,828,331	56.215	57.961
2026-10-14 / cc962159	679,449,939	653,831,656	88.546	90.149
2026-10-14 / 33816db5	679,549,997	654,627,509	88.344	90.874
2026-10-14 / 191a0739	679,240,609	653,664,759	85.135	87.917
2026-10-14 / cd226d0c	679,240,609	653,664,759	92.059	98.073
2026-10-14 / b2272a44	679,449,939	653,873,401	78.007	85.621

UTC date / revision	F4 Ir	F5 Ir	F4 ms	F5 ms
2026-10-14 / bc82d128	679,449,939	653,873,401	86.126	89.102
2026-10-14 / 912a98ed	679,098,623	653,830,968	89.995	91.125
2026-10-14 / 8d8b1295	679,098,623	653,830,968	89.395	90.230
2026-10-14 / b339e847	679,240,608	653,621,583	52.150	53.839
2026-10-15 / 7912faa5	679,226,318	653,606,027	88.406	88.091
2026-10-15 / 3c5d435e	678,905,208	653,631,022	90.855	91.956
2026-10-15 / f0599bf9	679,098,622	653,825,706	89.060	90.253
2026-10-15 / 3db0ecbc	679,449,938	653,830,913	86.813	90.492
2026-10-15 / addb6a02	679,256,525	653,632,330	54.248	56.951
2026-10-15 / 12348e50	679,449,939	651,714,930	89.150	90.941
2026-10-15 / b906df02	679,449,939	651,714,179	94.066	94.227
2026-10-15 / cf274521	679,549,984	654,627,503	88.061	90.795
2026-10-15 / 1b874fcc	679,449,939	653,830,975	86.246	88.831
2026-10-15 / 700f7d31	679,444,579	653,824,679	89.023	93.545
2026-10-15 / 289bb5fb	679,264,149	653,687,604	74.802	77.102
2026-10-16 / 92246530	679,449,938	653,829,966	75.128	75.009
2026-10-16 / 3a4a5cec	679,549,995	654,621,347	85.229	89.944
2026-10-16 / 333c08c0	672,605,656	647,427,799	76.341	78.831
2026-10-16 / 4cfcb94e	672,605,656	647,427,799	87.944	88.781
2026-10-16 / ca0310ff	672,486,486	649,202,105	89.012	87.501
2026-10-16 / de0c9391	672,500,776	649,216,408	74.498	77.487
2026-10-16 / 7051f4e2	679,449,938	653,873,407	76.675	78.834
2026-10-16 / 082d5507	679,445,034	653,868,507	71.580	80.148
2026-10-16 / 2f02e32c	679,549,984	654,627,503	85.711	90.622
2026-10-16 / 6d437f0c	679,549,984	654,627,503	51.852	54.190
2026-10-16 / 6bf2bcf0	679,549,997	654,621,861	90.619	91.074
2026-10-16 / f8300065	679,549,997	654,669,949	75.064	77.431
2026-10-16 / ccf43056	679,549,997	654,669,949	86.686	90.254
2026-10-16 / 57b028d8	679,350,531	654,421,882	84.863	88.297
2026-10-16 / f25d5e52	679,549,984	654,669,936	51.138	52.369
2026-10-16 / 213e7a80	679,449,939	653,830,971	85.272	87.253
2026-10-16 / 78adfda4	679,240,609	653,621,067	76.846	91.802
2026-10-17 / b1bf5b89	679,449,939	653,825,765	84.571	88.476
2026-10-17 / 40bcf690	679,449,939	653,825,765	80.639	80.305
2026-10-17 / 8efd4e48	679,549,984	654,622,293	88.930	89.875
2026-10-17 / 89036527	679,449,939	653,830,971	91.230	91.994
2026-10-17 / a90646f1	679,435,649	653,817,360	85.170	86.662
2026-10-17 / b59187e5	679,340,625	654,418,146	86.682	91.890
2026-10-17 / fd043d4f	679,549,955	654,621,012	90.548	92.367
2026-10-17 / 17c71374	679,449,939	653,830,397	86.043	89.344
2026-10-17 / 7d751c8d	679,098,491	654,182,814	55.523	58.269
2026-10-17 / a95fcd95	679,098,493	654,182,133	87.157	88.860
2026-10-17 / 2c1164df	679,198,639	690,798,036	88.066	93.755
2026-10-17 / e59342d3	679,198,639	690,798,036	85.604	94.177
2026-10-17 / ca00c151	679,198,639	690,798,036	88.918	96.426
2026-10-18 / f8501268	679,449,938	690,826,836	87.801	91.097
2026-10-18 / a1d0f1e6	679,449,938	690,826,836	88.118	93.700
2026-10-18 / c8613fe5	679,449,938	690,825,106	75.312	82.144
2026-10-18 / 3c192c34	679,449,938	690,825,106	93.514	100.589
2026-10-18 / 805b6279	679,549,993	691,623,334	89.125	96.021
2026-10-19 / 21d553b1	679,453,166	690,824,740	71.297	82.510
2026-10-19 / fc5d4835	679,449,937	690,826,048	92.362	98.482
2026-10-20 / 841a45cd	679,451,718	690,805,442	76.066	81.266
2026-10-20 / 0e3c51ed	679,340,663	691,396,123	55.582	62.124
2026-10-20 / d0337ff5	679,551,773	691,602,405	88.546	96.658
2026-10-20 / d30a63b8	679,549,994	691,600,618	92.571	95.491

Historical Python-orchestrated perf.history/v1 snapshots; no new performance run or speedup claim.

Runtime: one recorded median per kernel and revision; sample counts retained per point, no stored spread, no isolated timing eligibility. Runtime is diagnostic.

Profiler Ir is the recorded historical scope. Worker-thread coverage is incomplete for these kernels; it is not a complete solver operation count.

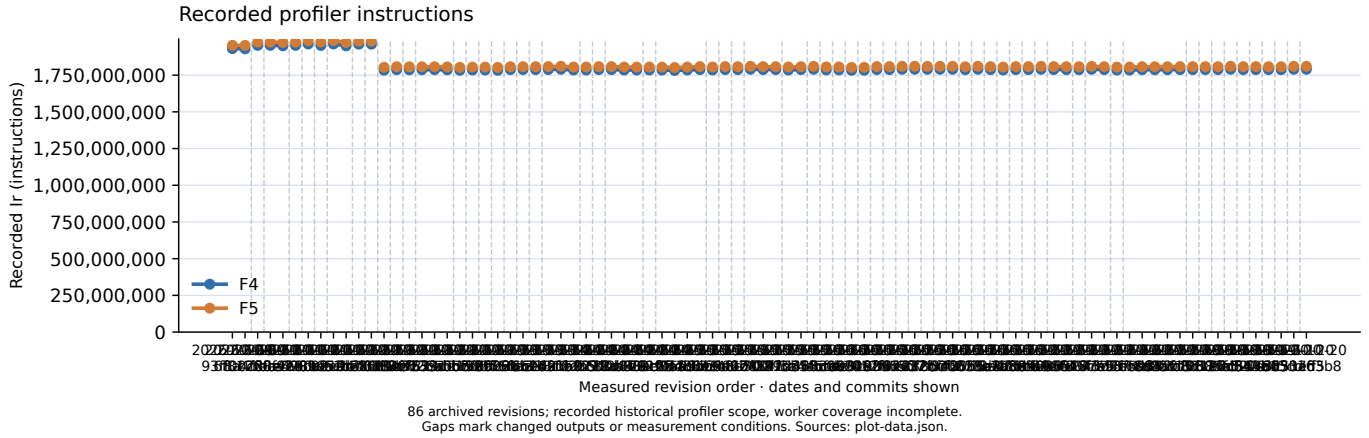
Lines stop at missing values, changed output fingerprints, or changed host/compiler/harness/profiler. Different algorithm fingerprints are expected and are not equated.

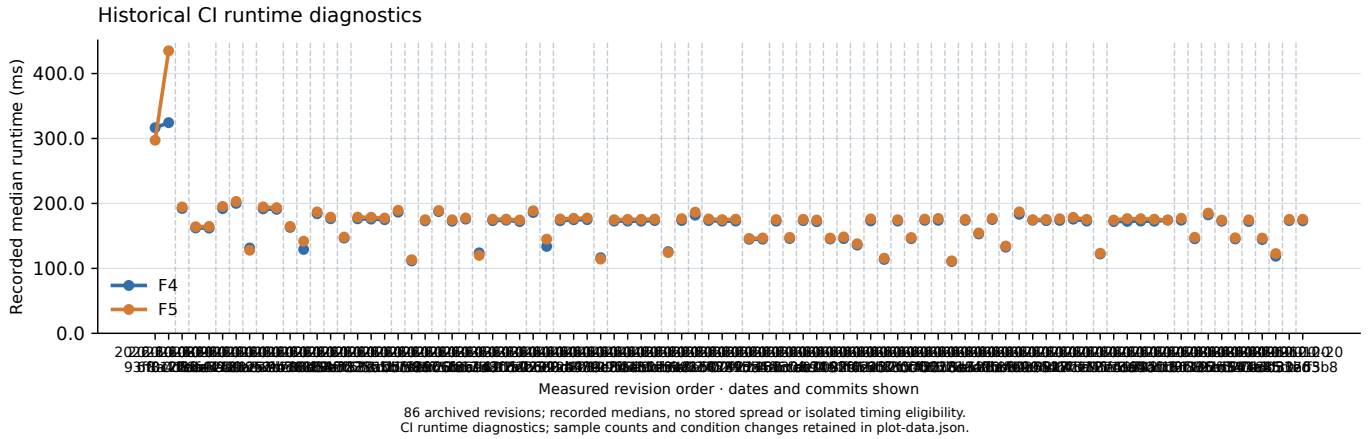
X positions are measured revision order, labelled with UTC commit date and revision; spacing is not elapsed time.

Evidence: 86 imported perf.history/v1 snapshots; historical_sources in plot-data.json retains raw filenames, SHA-256 hashes, hosts, harnesses and sample counts. Source data and provenance: plot-data.json and REPORT.md. These plots do not establish an end-to-end ECDLP gain.

Boolean solver calls: F4 and F5

Shared K_1 / 2^17 m=2 workload, 16 Boolean variables, four planted and four random targets, maximum degree 3. The solve_boolean_system entry point, including root readback.





UTC date / revision	F4 Ir	F5 Ir	F4 ms	F5 ms
2026-10-08 / 93ff0c72	1,930,989,200	1,953,108,992	316.570	297.297
2026-10-09 / 6f81260f	1,929,374,570	1,952,598,745	324.403	435.004
2026-10-11 / 8a41eca7	1,953,021,023	1,974,789,730	192.203	194.561
2026-10-11 / 0cfba411	1,953,020,766	1,975,267,516	162.266	164.161
2026-10-11 / f88e5909	1,949,981,977	1,973,219,339	161.982	164.584
2026-10-11 / fb926c02	1,953,020,766	1,974,789,781	191.988	195.374
2026-10-11 / e478b952	1,962,376,386	1,985,422,672	200.058	202.996
2026-10-11 / aa0cee7b	1,952,402,641	1,975,976,609	131.376	127.933
2026-10-12 / 81ca1c78	1,962,376,386	1,985,933,163	191.630	194.567
2026-10-12 / a8194d61	1,949,694,281	1,972,772,104	190.899	193.455
2026-10-12 / 5cbe16d5	1,961,124,313	1,981,687,200	163.135	164.619
2026-10-12 / eb70862e	1,961,124,313	1,981,686,883	129.169	141.847
2026-10-12 / 31088bfc	1,783,643,232	1,803,568,510	184.204	186.912
2026-10-12 / 0d34f782	1,787,314,371	1,807,797,314	176.437	178.824
2026-10-12 / 89e3233c	1,786,351,448	1,806,842,750	146.640	148.159
2026-10-12 / 80f5391d	1,787,314,371	1,807,244,283	176.353	178.878
2026-10-12 / c62555b7	1,787,314,371	1,807,244,283	175.899	178.689
2026-10-12 / 633d2158	1,786,234,678	1,806,168,901	174.990	177.494
2026-10-12 / ca0d5f86	1,783,289,117	1,802,994,551	186.580	189.492
2026-10-12 / b5cc6ffc	1,785,257,434	1,804,956,249	111.497	113.333
2026-10-13 / bf92906b	1,785,257,434	1,804,956,249	173.194	174.965
2026-10-13 / d85756aa	1,783,003,746	1,802,715,389	187.409	189.039
2026-10-13 / ab28b57b	1,787,965,641	1,807,670,514	172.579	174.785
2026-10-13 / 56f66e43	1,787,969,850	1,807,674,723	175.850	177.627
2026-10-13 / 28b1daf2	1,787,969,850	1,807,674,723	123.983	119.998
2026-10-14 / cc962159	1,787,965,641	1,808,168,023	173.510	175.690
2026-10-14 / 33816db5	1,790,194,799	1,810,456,555	174.166	175.821
2026-10-14 / 191a0739	1,785,256,826	1,805,459,208	172.076	174.475
2026-10-14 / cd226d0c	1,785,256,826	1,805,459,208	185.930	188.735
2026-10-14 / b2272a44	1,787,965,186	1,808,230,025	133.809	144.964

UTC date / revision	F4 Ir	F5 Ir	F4 ms	F5 ms
2026-10-14 / bc82d128	1,787,965,186	1,808,230,025	173.338	176.053
2026-10-14 / 912a98ed	1,784,912,427	1,805,177,722	174.838	177.121
2026-10-14 / 8d8b1295	1,784,912,427	1,805,177,722	175.021	177.613
2026-10-14 / b339e847	1,785,255,750	1,805,524,217	116.412	114.358
2026-10-15 / 7912faa5	1,784,970,955	1,804,687,873	172.752	174.937
2026-10-15 / 3c5d435e	1,782,202,991	1,801,914,239	172.763	175.551
2026-10-15 / f0599bf9	1,784,990,781	1,805,255,060	172.667	175.531
2026-10-15 / 3db0ecbc	1,787,964,110	1,808,232,577	173.668	175.897
2026-10-15 / addb6a02	1,785,268,499	1,805,533,338	125.807	124.378
2026-10-15 / 12348e50	1,788,056,225	1,808,318,933	173.678	176.547
2026-10-15 / b906df02	1,788,056,225	1,807,750,349	181.927	186.574
2026-10-15 / cf274521	1,790,285,878	1,810,547,645	173.613	176.055
2026-10-15 / 1b874fcc	1,787,976,859	1,808,241,709	172.732	175.034
2026-10-15 / 700f7d31	1,787,976,859	1,807,671,995	172.808	175.713
2026-10-15 / 289bb5fb	1,785,371,563	1,805,638,717	144.794	146.086
2026-10-16 / 92246530	1,788,058,429	1,807,748,501	144.989	146.889
2026-10-16 / 3a4a5cec	1,790,297,890	1,809,989,439	172.711	175.194
2026-10-16 / 333c08c0	1,785,727,004	1,805,891,517	146.016	147.750
2026-10-16 / 4cfcb94e	1,785,727,004	1,805,891,517	173.541	175.495
2026-10-16 / ca0310ff	1,783,097,959	1,802,805,840	171.952	174.458
2026-10-16 / de0c9391	1,783,382,754	1,803,084,509	145.240	146.698
2026-10-16 / 7051f4e2	1,788,058,429	1,808,325,583	145.816	148.406
2026-10-16 / 082d5507	1,788,058,429	1,808,325,583	135.905	137.873
2026-10-16 / 2f02e32c	1,790,289,796	1,810,551,563	173.111	176.139
2026-10-16 / 6d437f0c	1,790,289,796	1,810,551,563	113.741	115.682
2026-10-16 / 6bf2bcf0	1,790,220,548	1,809,884,826	172.852	174.962
2026-10-16 / f8300065	1,790,220,548	1,810,482,315	145.826	147.493
2026-10-16 / ccf43056	1,790,220,548	1,810,482,315	173.926	175.928
2026-10-16 / 57b028d8	1,787,512,707	1,807,204,760	173.993	176.851
2026-10-16 / f25d5e52	1,790,210,949	1,810,472,716	110.513	111.595
2026-10-16 / 213e7a80	1,787,980,594	1,808,241,908	173.897	175.384
2026-10-16 / 78adfda4	1,785,272,234	1,804,971,060	152.949	154.518
2026-10-17 / b1bf5b89	1,788,059,441	1,808,326,595	175.225	176.824
2026-10-17 / 40bcf690	1,788,059,441	1,808,326,595	132.645	134.442
2026-10-17 / 8efd4e48	1,790,210,430	1,810,472,197	183.238	186.980
2026-10-17 / 89036527	1,788,059,441	1,808,322,547	174.065	174.959
2026-10-17 / a90646f1	1,787,695,280	1,807,903,799	173.534	175.412
2026-10-17 / b59187e5	1,787,474,342	1,807,737,121	173.891	176.406
2026-10-17 / fd043d4f	1,790,182,702	1,809,892,450	175.761	178.537
2026-10-17 / 17c71374	1,787,980,075	1,807,678,901	172.788	175.501
2026-10-17 / 7d751c8d	1,784,810,930	1,805,015,067	122.130	123.306
2026-10-17 / a95fcd95	1,784,812,954	1,805,078,260	172.140	174.437
2026-10-17 / 2c1164df	1,787,129,943	1,807,393,178	172.286	176.688
2026-10-17 / e59342d3	1,787,129,943	1,807,393,178	172.825	176.495
2026-10-17 / ca00c151	1,787,129,943	1,807,393,178	172.532	175.739
2026-10-18 / f8501268	1,787,980,091	1,808,180,618	174.301	174.497
2026-10-18 / a1d0f1e6	1,787,980,091	1,808,180,618	174.167	177.133
2026-10-18 / c8613fe5	1,787,980,091	1,807,674,211	145.609	147.871
2026-10-18 / 3c192c34	1,787,980,091	1,807,674,211	182.414	185.143
2026-10-18 / 805b6279	1,790,216,532	1,810,414,822	172.598	174.347
2026-10-19 / 21d553b1	1,787,695,296	1,807,902,799	145.303	147.070
2026-10-19 / fc5d4835	1,787,979,079	1,808,242,913	172.170	174.760
2026-10-20 / 841a45cd	1,787,979,207	1,807,773,273	144.351	146.669
2026-10-20 / 0e3c51ed	1,787,512,766	1,807,304,761	118.682	122.765
2026-10-20 / d0337ff5	1,790,216,532	1,809,990,751	173.574	175.443
2026-10-20 / d30a63b8	1,790,222,138	1,809,996,357	173.214	175.560

Historical Python-orchestrated perf.history/v1 snapshots; no new performance run or speedup claim.

Runtime: one recorded median per kernel and revision; sample counts retained per point, no stored spread, no isolated timing eligibility. Runtime is diagnostic.

Profiler Ir is the recorded historical scope. Worker-thread coverage is incomplete for these kernels; it is not a complete solver operation count.

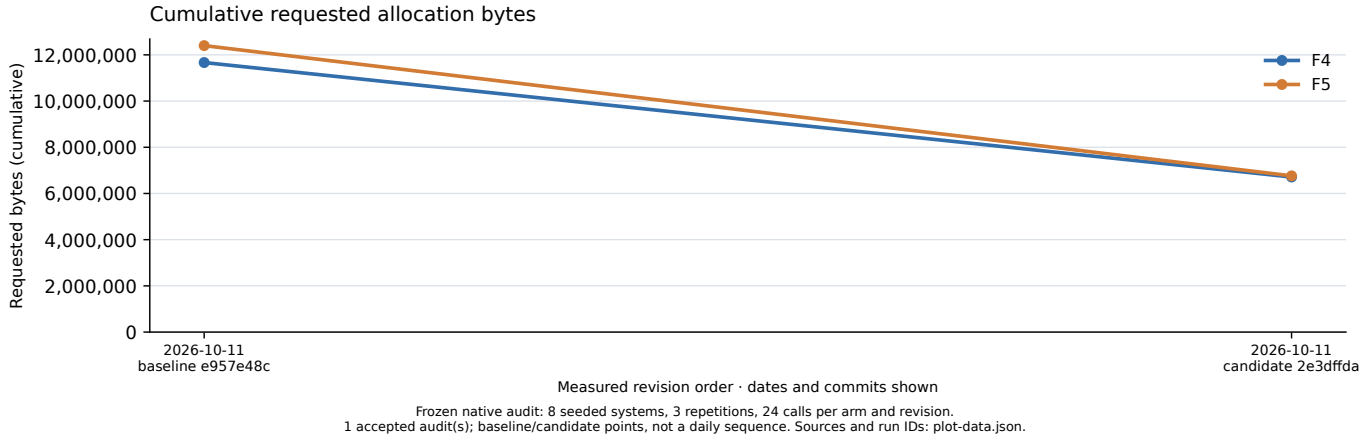
Lines stop at missing values, changed output fingerprints, or changed host/compiler/harness/profiler. Different algorithm fingerprints are expected and are not equated.

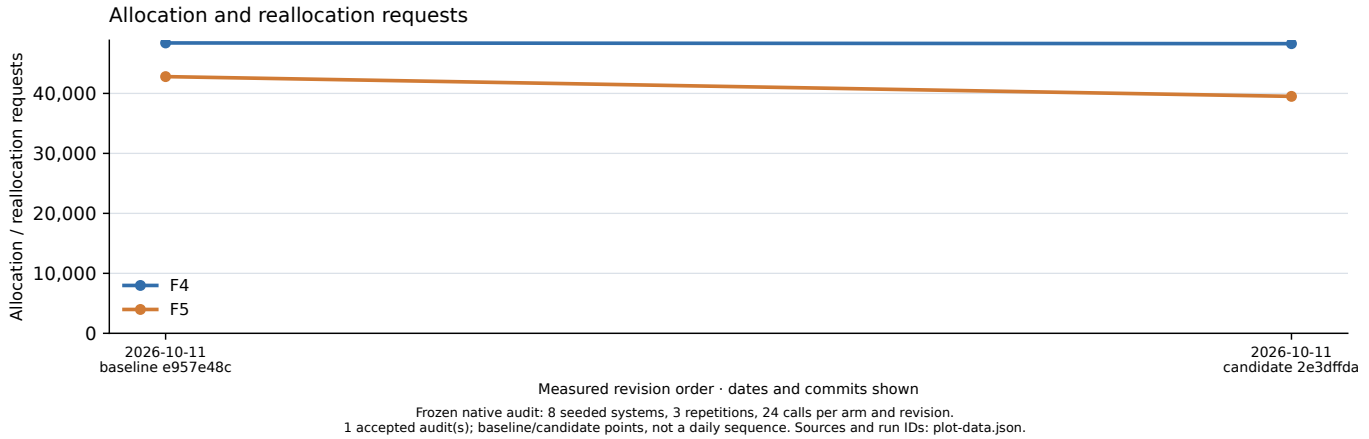
X positions are measured revision order, labelled with UTC commit date and revision; spacing is not elapsed time.

Evidence: 86 imported perf.history/v1 snapshots; historical_sources in plot-data.json retains raw filenames, SHA-256 hashes, hosts, harnesses and sample counts. Source data and provenance: plot-data.json and REPORT.md. These plots do not establish an end-to-end ECDLP gain.

Complete Boolean calls: allocation history

Matching frozen Boolean allocation suite: 8 seeded systems, 3 repetitions, 24 calls per arm and revision. Polynomial dimensions, degrees and seeds are retained in the source audit protocol.





UTC date / revision	F4 bytes	F5 bytes	F4 requests	F5 requests
2026-10-11 / e957e48c	11,667,696	12,400,920	48,414	42,801
2026-10-11 / 2e3dffda	6,712,608	6,762,696	48,294	39,507

Native allocation audit. Original baseline/candidate rows, accepted comparison, exact outputs and accounting, shared inputs and all totals revalidated by the Rust plot-data tool.

One baseline and one candidate point per accepted audit, not a daily sequence. A/A and replication pairs are controls, not additional algorithm revisions.

Bytes are cumulative requested traffic, not peak memory. No timing is measured by this allocation series.

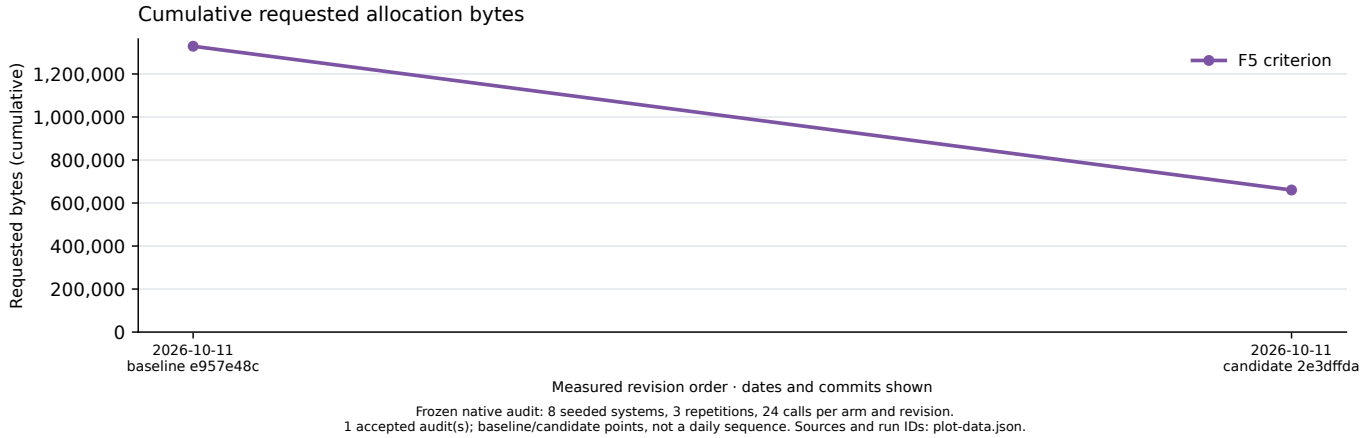
Criterion construction is a separate part of F5 and is never compared as a complete solver.

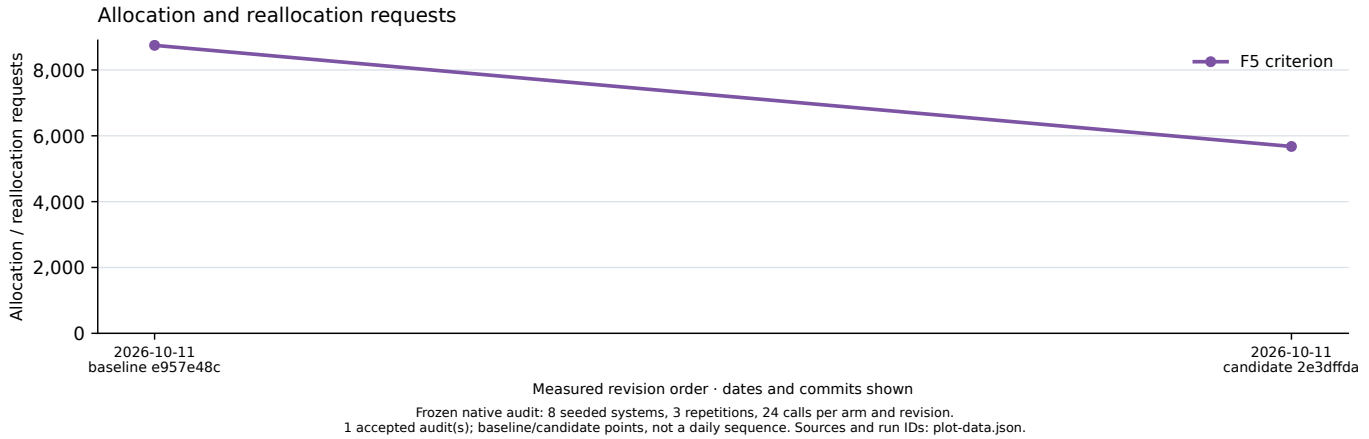
Audit revisions come from their frozen records. They are not inserted as measurements of historical CI workloads; the workloads and metrics remain separate.

Evidence: allocation_sources in plot-data.json; frozen native comparison and baseline/candidate JSONL records in research/f4_f5_row_build_20261011/records/. Source data and provenance: plot-data.json and REPORT.md. These plots do not establish an end-to-end ECDLP gain.

F5 criterion: allocation history

Matching frozen Boolean allocation suite: 8 seeded systems, 3 repetitions, 24 calls per arm and revision. Polynomial dimensions, degrees and seeds are retained in the source audit protocol.





UTC date / revision	F5 criterion bytes	F5 criterion requests
2026-10-11 / e957e48c	1,329,024	8,748
2026-10-11 / 2e3dffda	660,480	5,676

Native allocation audit. Original baseline/candidate rows, accepted comparison, exact outputs and accounting, shared inputs and all totals revalidated by the Rust plot-data tool.

One baseline and one candidate point per accepted audit, not a daily sequence. A/A and replication pairs are controls, not additional algorithm revisions.

Bytes are cumulative requested traffic, not peak memory. No timing is measured by this allocation series.

Criterion construction is a separate part of F5 and is never compared as a complete solver.

Audit revisions come from their frozen records. They are not inserted as measurements of historical CI workloads; the workloads and metrics remain separate.

Evidence: allocation_sources in plot-data.json; frozen native comparison and baseline/candidate JSONL records in research/f4_f5_row_build_20261011/records/. Source data and provenance: plot-data.json and REPORT.md. These plots do not establish an end-to-end ECDLP gain.